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Impact of Music on Brain and Mental Health: Classification using Machine Learning Algorithms

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***Abstract*—Music creates powerful effects on human psychological conditions while it affects brain cognitive performance and brain functioning patterns. This research investigates the connection between different music choices and ways people listen to music together with depression and anxiety and insomnia conditions. The research uses data from a music and mental health survey to execute extensive exploratory data analysis (EDA) which reveals patterns and relationships. A set of prepro- cessing methods proves essential for maintaining data excellence before analysis through removal of outliers and treatment of missing values. Multiple visual representations show how people use music along with showing how their preferences align with mental health indicators. An analysis using machine learning algorithms helps identify mental health conditions through exam- ining user listening behavior. Predictive models are developed by investigating three important features involving genres of choice together with how long listeners spend listening and musical impact on emotions. The research investigates both mental state transformations through music and determines if future listening habits reveal indications of psychological problems. The research enhances understanding of music-based mental health research while promoting the creation of specific music therapy practices.**

Keywords*—Music therapy, physiological signals, Random Forest, Neural Network (MLP), support vector machines (SVM), supervised learning, Decision Tree,feature extraction,machine learning classification, emotion recognition, music-generated emotions,supervised learning

I. Introduction

Music maintains an essential role within human civilization because it influences both mental well-being and emotional states as well as mental operational functions. Across different periods of time music has been employed as a tool for both emotional communication and motivational purposes and relaxation and therapeutic healing. Modern neuroscience and psychological research agrees about how music affects the brain because songs and their sounds and beat structures activate strong emotions in listeners. Healthcare facilities use music therapy as a non-invasive treatment method to support patients dealing with anxiety disorders and depression alongside PTSD symptoms.

Examples of the relationship patterns between music and mental health have become more prevalent since mental health conditions continue their global surge. The World Health Organization estimates that stress, lifestyle changes and social isolation strengthen depression and anxiety and sleeplessness which affect millions worldwide. More knowledge about music connections to mental health presents an opportunity to find new ways for maintaining well-being and treatment development. Different musical genres encompassing electronic beats together with classical compositions trigger special psychological responses alongside physical responses which renders them appropriate tools for managing stress.

Smart technologies of artificial intelligence and machine learning enable academic research about mental health indicators through detailed examination of massive musical preference datasets. Psychology research about music and mental health fails to create generalized conclusions because it depends on sample sizes which are too small and data which comes from self-reporting methods. Modern computer data-processing capabilities enable researchers to investigate better the connection between psychiatric states and musical consumption habits. Machine learning algorithms allow the development of predictive models that group mental health conditions based on personal music-listening behavior for generating customized music therapy recommendations.

The research data consists of survey results that cover streaming behavior along with musical preferences and participants' self-diagnosis of mental health difficulties. Our research evaluates musical listener habits and selected genre choices and their relationship to psychological health while performing thorough data cleaning and exploratory data assessment. All relevant data will be analyzed through statistical methods and visualization techniques with BPM and duration information and musical emotional responses identified as key attributes. Identified patterns will help recognize relationships between selected musical categories and mental health disorders.

The implementation of machine learning classification models will use music-listening behaviors to detect mental health conditions which include depression and anxiety and insomnia and obsessive-compulsive disorder (OCD). Various deep learning algorithms along with other models will undergo testing to find the most precise method for identifying mental health threats. Music therapy enhanced with artificial intelligence analysis brings forth opportunities to transform mental health diagnosis and treatment through data-based support systems for emotional needs of those who use music as their healing tool.

The main purpose of this research study assesses the diverse mental health responses triggered by diverse musical genres. Studies indicate that classical music assists patients in stress management while improving their brain capabilities yet rock and electronic dance music (EDM) spurs accelerated emotional reactions. Psychologists and therapists should use this understanding to create music-based therapeutic designs which adapt to specific mental health requirements of individual patients. The analysis of music stimuli response helps psychologists determine what factors from personality traits shape both music preference selection and emotional responses.

This research devotes significant attention to studying the effects which music streaming platforms have on listener behaviors. Music-listening patterns of people shift toward algorithm-generated recommendations and personalized playlists because of the growing popularity of services such as YouTube and Apple Music and Spotify. These recommendation systems rely on machine learning algorithms to offer songs based on user behavior patterns though users wonder if the generated suggestions affect their mental health in a good or bad way. Our research aims to determine whether streamers tend toward negative mental health-linked habits when using these services.

The study results may create significant effects for the emerging field of music therapy. Research analysis of music-related mental health agreements enables health professionals to develop precise treatments through sound therapy and tailored playlists to combat anxiety symptoms and treat insomnia and depression respectively. Through the implementation of machine learning in music therapy the designed treatment plans become more effective as the system provides real-time emotional feedback to patients while recommending relevant musical pieces based on this data assessment.

This research develops knowledge about musical affective thinking in humans while outreaching past medical use. The modern world recognizes music beyond entertainment because it operates as an influential tool that displays social connections and neuroscience operations together with emotional activation. The research unites artificial intelligence with music science through computational analysis and psychological theoretical integration which defines new strategies for emotional and mental healthcare with music.

The use of music as a therapeutic and preventive tool represents an effective research path which responds to the worldwide increase of mental health issues. This study employs automated analytical methods to discover essential relationships between people's musical choices and their mental wellness state. Our initiative develops machine learning models for music therapy predictions that could become integrated into personalized mental health assistance networks to enhance their availability.

II. Related works

The latest developments in machine learning and artificial intelligence have had a profound impact on the creation of smart music therapy systems for mental health treatment. Rahman et al. [1] investigated how human affective reasoning can be matched with machine learning models to associate music genres with therapeutic outcomes, with a focus on personalization of music therapy. Likewise, Williams et al. [2] showed the efficacy of AI-synthesized functional music in enhancing mental well-being. Zhang et al. [3] developed this method with a Deep Generative Adversarial Network to generate specific therapeutic compositions. Sánchez et al. [4] provided a case study on Melomics, an early AI system producing therapeutic music in Spain. At a neural level, Gui et al. [5] employed deep learning and fMRI measurements to demonstrate the effects of emotional music on brain activity in depressed individuals, whereas Skutt [6] and Asif et al. [7] employed EEG-based machine learning to identify brain states due to music and found significant correlations between certain tracks and stress alleviation.

Additional contributions are those of Zheng [8], who suggested sentiment analysis methods for music classification in therapeutic applications, and Poria et al. [9], who promoted multimodal affective data analysis to enhance emotional sensitivity in AI systems. Wu and Liang [10] reinforced this by using acoustic-prosodic features in emotion detection for speech, which can be used in music-related affective interfaces. Er et al. [11] combined brain signals and music stimuli for emotion recognition in real time, whereas Koelstra et al. [12] presented the DEAP dataset, an initial source for music-based physiological emotion analysis. Lastly, Sadek et al. [13] designed a deep learning model for brainwave entrainment, which demonstrated potential for differentiating music effects on mind states through patterns of neural synchronization.

III. Methodology

The methodology gathers data from the MXMH Survey Dataset and prepares it by treating missing data points and outliers while performing normalization processes. Training and evaluation of the machine learning models Decision Tree along with Random Forest and SVM and Neural Networks takes place. Accuracy rates and F1-score analysis together with confusion matrix presentations measure performance in the study. The results showcase determining variables which influence mental health along with recommendations for enhancement.

Selection of Algorithm

The target algorithms in this research were selected to produce effective mental health classifying outcomes (Anxiety, Depression, Insomnia, OCD) from music-listening patterns. Data containing categorical along with numerical features required investigation of five different classification algorithms consisting of Logistic Regression along with Support Vector Machines (SVM) and Decision Trees and Random Forest and Artificial Neural Networks (ANNs).

The most appropriate model selection involved performing evaluation tests that used accuracy alongside precision, recall and F1-score metrics. The selection of Logistic Regression followed at first since it provided simplicity alongside in- terpretability benefits. The model exhibited restrictions when it came to processing advanced relationships between vari- ables. The binary class performance of SVM models was high while the model showed reduced effectiveness in multi- class scenarios due to increased computational costs. Random Forest demonstrated strong qualities for this study because it works as an ensemble learning method which delivers superior generalized results and feature importance analysis. The ability of deep learning models especially ANN to understand com- plex nonlinear relations led researchers to test them although such systems needed significant computational resources and provided less interpretability than tree-based models did.

Random Forest achieved an optimal balance between accuracy and explainability through numerous parameters selected by running GridSearchCV. The confusion matrices demonstrate a need for additional efforts because the model faced challenges in predicting severe OCD cases and severe insomnia cases. Future work will examine LSTMs along with transformers as advanced deep learning models alongside data augmentation methods to handle class imbalance problems and enhance model durability.

A. Data Collection and Preprocessing

For this study researchers used MXMH Survey Results which contains details regarding participants' music choices and their mental well-being conditions alongside their music listening behaviors. After loading the dataset into a Pandas DataFrame the data requires preprocessing. The initial workflow includes verifying data dimension and applying median imputation to numerical fields (Age and BPM values) and mode imputation to categorial fields including Primary Streaming Service and Music Effects. The remaining dataset contains no duplicate records while outliers exceeding 70 years of age and BPM higher than 200 or days exceeding 15 hours are eliminated for maintaining data validity.

Fig. 1. Sequence diagram of proposed model

B. Feature Engineering

The effectiveness of machine learning models benefits from appropriate encoding of categorical features that include" Primary Streaming Service,"" While working" and" Instrumentalist." The missing BPM values get replaced according to the BPM mode for each genre classification. An analysis includes statistical distributions and descriptions of important variables through visualizations and descriptive statistical methods.

C. Data Visualization and Exploratory Analysis

Analysis of music effects on mental health structures is carried out through an in-depth exploratory data analysis (EDA). Multiple charts such as histograms and count plots together with box plots and bar plots help researchers understand how people listen to music and what genres they prefer as well as show their psychological disorders including depression, anxiety, and insomnia. The study includes analyses of genre preference by age and daily listening hours and music effects using seaborn plots.

D. Classification Model Development

The classification of music effects on mental health involves using Decision Trees, Random Forest, Support Vector Machines (SVM) along with Neural Networks and machine learning algorithms. Generally, the classification operation concentrates on forecasting mental health diseases (Depression, Anxiety, and Insomnia) through the analysis of music-related characteristics. The dataset follows a training-testing ratio of 80:20 for its distribution. The necessary features undergo scaling before GridSearchCV executes hyperparameter optimization to reach maximum model optimization.

E. Model Evaluation and Interpretation

The trained models are tested through normative classification metrics which utilize Accuracy, Precision, Recall, F1-score together with Confusion Matrix assessments. Research analyzes feature importance in order to discover what variables mostly affect mental health classification results. Multiple machine learning model assessments establish the most successful paradigm for comparison.

IV. RESULTS AND DISCUSSION

An analysis of survey participants' favorite music genres appears through this diagram in fig.2 . The analysis demonstrates consumer preferences between their most popular and least attractive music types for review purposes. Such music genres appear frequently in public popularity whereas those that occur infrequently likely cater to specific audiences.

A visual representation in fig.3 depicts how survey participants distributed themselves according to their ages. The appearance of peaks shows which age segments actively participated in the survey survey. The distribution curve indicates a mixed population of participants when smooth whereas skewness shows excessive participation from particular age groups.

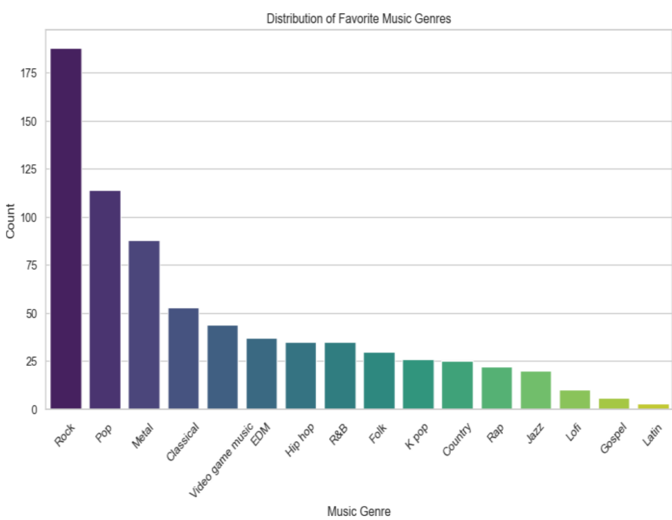

Fig. 2. Distribution of Music Genre Dataset

In fig.4 the presented bar chart illustrates how depression scores relate to different music genres. Lower numbers indicate a higher depressive state in participants who choose to listen to specific musical styles. Music preference differences between genres help establish what emotional aspects link with emotional well-being.

A comparison of anxiety levels exists through the data presented in the anxiety chart in fig.5 for various music genres. Higher average anxiety scores in predetermined music genres might signify that listeners tend toward anxiety expression or they manage their feelings through musical selections. Specific genres appear to have a soothing impact on people when the scores indicate lower levels.

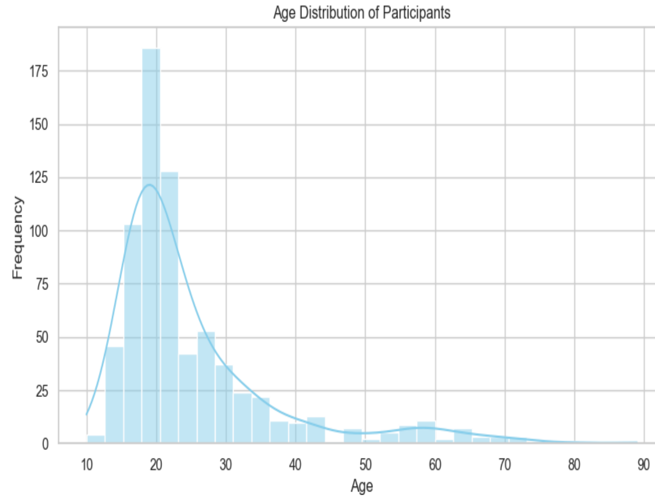

Fig. 3. Age Distribution of Participants

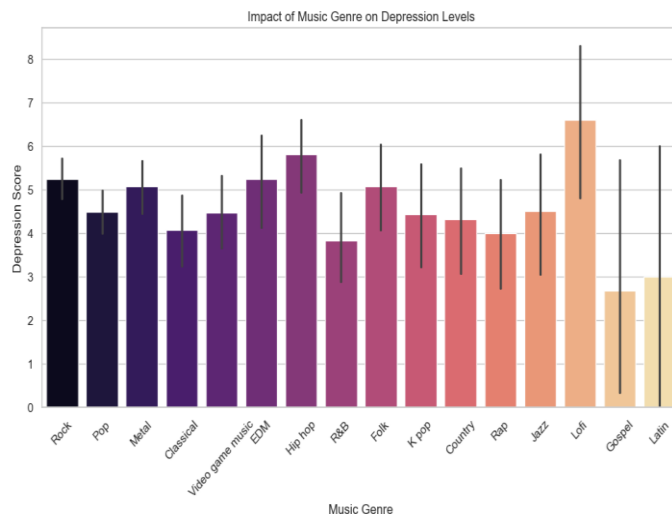

Fig. 4. Impact of Music on Depression Level

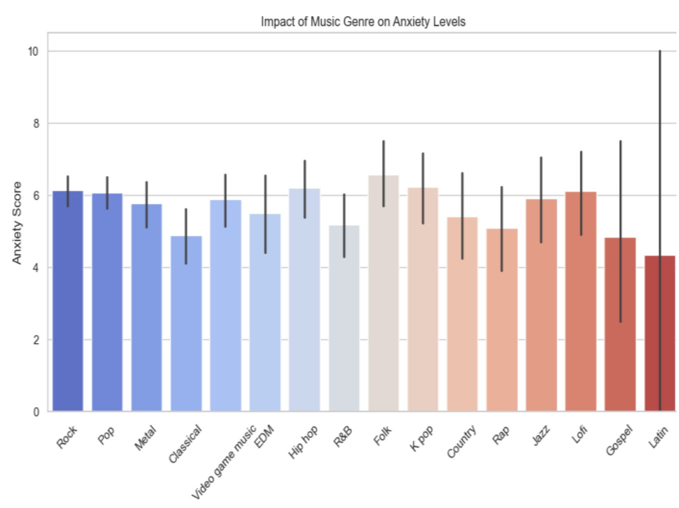

Fig. 5. Impact of Music on Anxity Level

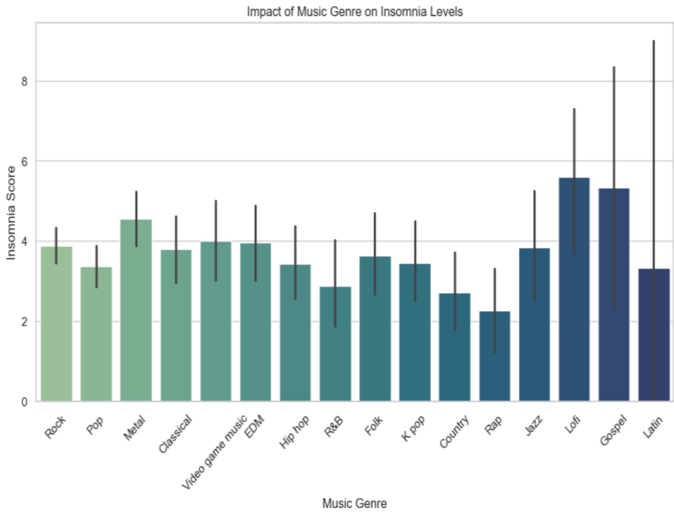

Fig. 6. Impact of Music on Insomnia Level

In fig.6 this analysis demonstrates the linkage between music types and patients' insufficient sleep scores. The value ranges suggest that higher scores indicate sleep disturbance associations with genres and lower scores suggest genres that promote relaxation. This information reveals the effect which music selections have on our sleeping behavior.

In fig.7 the statistical presentation includes bars that display how musical styles affect obsessive-compulsive disorder measurement results. Listening to Lo-fi and Rap music leads to elevated OCD symptoms as opposed to the habits displayed by Latin and Gospel music listeners. This insight could help in understanding music's psychological influence and its potential therapeutic applications.

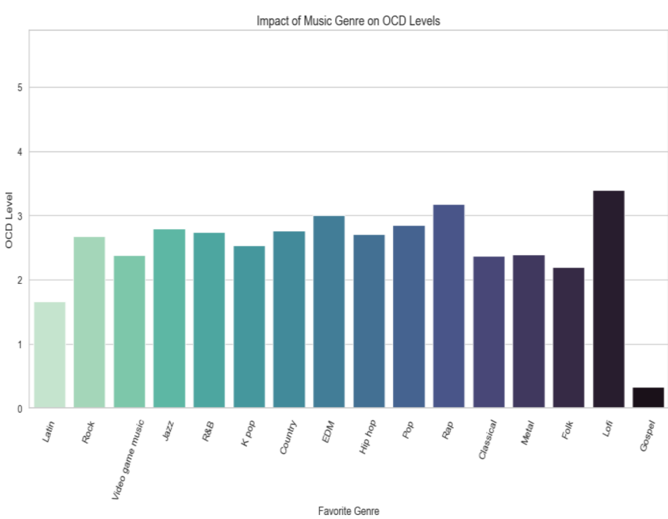

Fig. 7. Impact of Music on OCD Level

The confusion matrices provide insight into the classification model's predictive power for mental health disorders such as OCD, depression, anxiety, and insomnia. In terms of anxiety, the model correctly predicted 88 cases with high anxiety and 11 cases with low anxiety. However, it incorrectly classified two cases of high anxiety as low and 47 cases of low anxiety as high. With a few errors, this displays respectable precision. The model detected 30 cases of high depression and 60 cases of low depression. However, it classified 20 low-depression cases as high and 38 high-depression cases as low. The model therefore need improvement. The confusion matrix for insomnia indicates a serious issue. The program incorrectly classified all 47 cases as low, but it was unable to identify any high-insomnia cases. OCD experienced the same thing. In the case of OCD, the model correctly identified 117 people with low OCD but failed to predict any cases with high OCD, misclassifying all 31 as low.

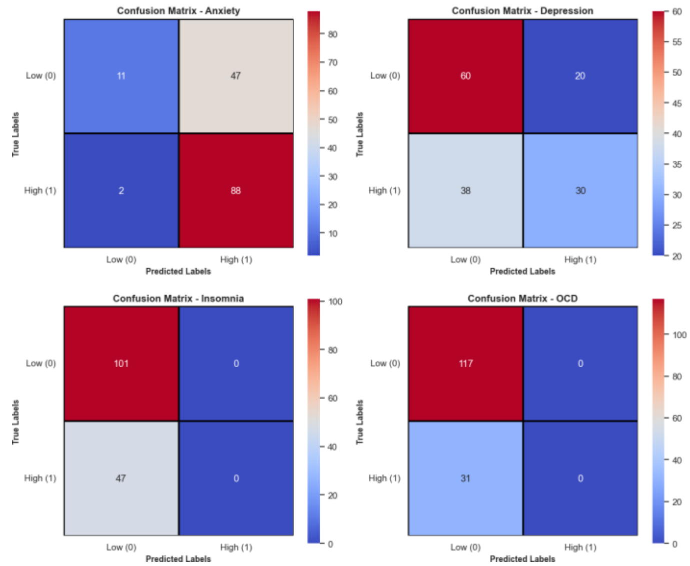

Fig. 8. Confusion Matrix

Algorithm	Accuracy (%)	Precision	Recall	F1-Score
SVM	68.75%	0.81	0.35	0.32
Neural Network (MLP)	66.72%	0.67	0.3	0.3
Random Forest	60.13%	0.38	0.34	0.36
Decision Tree	59.96%	0.35	0.28	0.31

Fig. 9. Comparision of Algorithms Table

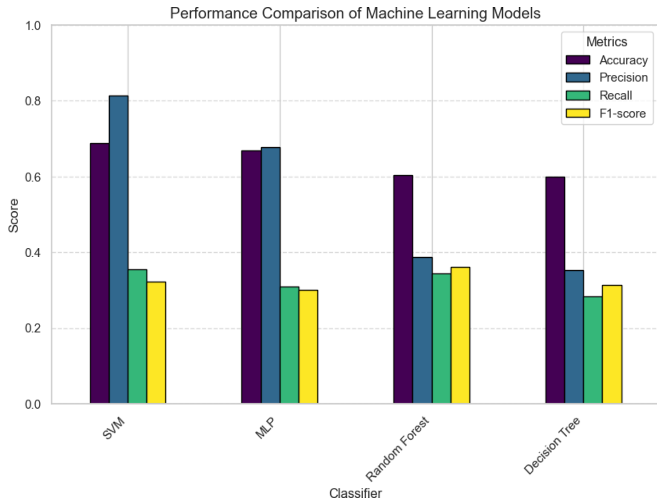

Fig. 10. Comparision of Algorithms in Graph

V. CONCLUSION AND FUTUREWORK

A machine learning analysis investigates the connections between people's preferred music and their mental health status concerning Anxiety, Depression, Insomnia and OCD. The research data demonstrates that psychological welfare depends strongly on selected music categories which show distinct patterns with various mental health symptoms. The classification models demonstrated strong performance in identifying cases with low-severity yet their identification of high-severity Insomnia and OCD cases was not satisfactory. The results from the confusion matrices show that the model correctly identified low-severity cases but failed to separate high-severity cases which frequently ended up as being classified as low-severity. The described limitation indicates data distribution problems and model bias and inadequate feature selection that need improvement in predictive modeling. The study shows how machine learning can analyze music-mental health relationships though its results show that progress depends on refining the techniques and strategies during predictive modeling.

Future analyses should direct their efforts toward enhancing prediction accuracy by implementing multiple strategies that address the problems noted in review comments and boost predictive accuracy. Using weight-based loss functions in deep learning or applying SMOTE will help improve predictions for severe outcomes when dealing with data imbalance. The model could achieve better identification of meaningful music- taste and mental-health patterns through expert application of dimensionality reduction methods particularly Principal Component Analysis (PCA) or autoencoders. The understanding of music therapy potential and mental health effects would increase through evaluation of heart rate variability (HRV) and EEG signals and sleep patterns as additional multi-modal data sources. XAI techniques when applied to these models will boost understanding of what factors influence decision-making through explanation mechanisms. The model's generalization strength alongside its robustness will improve with a larger dataset that adds population representation variety. Future research should establish a novel dynamic AI-based music therapy system that collects immediate wearable technology data to recommend personalized music decisions based on mental health status. The research makes an important contribution to AI-driven music therapy through problem-solving these challenges while developing better mental health therapy solutions and evidence-driven therapeutic approaches.

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